

Learning Bounded Latent Degradation Dynamics for stable rollout and Remaining Useful Life Prediction

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ABSTRACT

Long-horizon multivariate time-series forecasting is often unstable when models are recursively rolled out in observation space. This problem is especially pronounced in degradation modelling, where sensor trajectories are noisy, regime-dependent, and only partially informative about latent health. We study whether low-dimensional bounded latent representations can act as dynamical regularizers for stable rollout. The proposed algorithm identifies operating regimes from control variables, normalizes sensor measurements within regimes, denoises the resulting trajectories, selects degradation-sensitive channels, and trains a bounded latent autoencoder with monotonicity and smoothness regularization on a primary health coordinate. We provide a theoretical analysis separating sensor-space closed-loop divergence, bounded latent rollout, and smoothness-controlled forecastability. Experiments on NASA C-MAPSS show that higher-dimensional latent spaces can improve reconstruction while weakening rollout stability. Ablations further show that finite-horizon decoded boundedness is not unique to the bounded latent model: an unbounded autoencoder may also remain numerically bounded under constant-velocity rollout because of small latent velocities and decoder saturation. The main empirical advantage of the proposed representation is therefore forecastability: lower latent curvature, higher forecast skill, and useful downstream remaining-useful-life prediction.

1. Introduction

Many multivariate time-series applications require models that remain useful when rolled forward beyond a one-step training objective. In degradation modelling and prognostics, this requirement is especially important: forecasts are used not only to interpolate observed measurements, but also to reason about future system health. A model that reconstructs historical trajectories accurately may still produce implausible or unstable behavior when recursively rolled out.

Raw sensor measurements mix several sources of variation. Measurements change with operating conditions, sensor noise, unit-specific behavior, and latent degradation. Directly forecasting in observation space can therefore lead to unstable dynamics. Linear autoregressive models may become non-contractive when their closed-loop transition operators have spectral radius greater than one (Hamilton, 1994; Lütkepohl, 2005), while high-capacity sequence models may achieve good short-horizon accuracy without guaranteeing stable free-run behavior, a difficulty long documented for recurrent architectures trained via backpropagation through time (Pascanu, Mikolov and Bengio, 2013).

This paper studies an algorithmic alternative. Instead of forecasting raw sensors directly, we learn a compact latent state intended to represent degradation after removing operating-regime effects. The latent state is bounded by construction and is regularized so that its primary coordinate evolves smoothly and approximately monotonically over the life of each unit. Forecasting is then performed by local constant-velocity rollout in latent space, followed by decoding back to the sensor representation.

The key thesis is that stable long-horizon forecasting requires more than accurate reconstruction. Bounded latent geometry constrains the admissible state space, while smooth low-curvature latent dynamics make motion within that space forecastable. These are distinct mechanisms: boundedness alone is not sufficient for useful forecasting, and short-horizon accuracy does not ensure stable free-run behavior.

The main contributions are:

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